HISAABSET (LedgerLoop)

B2B Receivables Decision Desk & Automated Collections Engine

NMIMS FyndHacks 2026 — Track C28 Official Submission Package

Submission Track:	**C28: Receivables Follow-Up Desk & Retailer Credit Decision Engine**
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Team Members & Roles:	**Faiz Sayeed** (Frontend UX & OCR) | **Sai Kumar Reddy** (Financial Logic & QA)
Public GitHub Repository:	https://github.com/MdZeeshanAli01/LedgerLoop (Branch: main)
Exact Commit SHA:	41909b15acb83751a25e296a40981ad7e8547374 (Verified clean HEAD)
24/7 Live AWS Deployment:	**http://13.201.115.214:8501** (AWS ECS on Fargate, ap-south-1 Mumbai)
Demo Credentials:	Role: **Finance Lead** | User: admin | Password: hisaabset123 (Quick Login enabled)
Automated Test Suite:	**89 Tests Passed** (100% Core Rules, Loader, Disputes, Payments, SES dispatch)

Executive Venture Summary

In India's unorganized supply chains, over 350,000 wholesale distributors (FMCG, pharmaceuticals, hardware, paints) extend short-term trade credit to local kirana retailers. These distributors routinely suffer from **delayed payments, artificial dispute claims, and frozen working capital**. While traditional accounting packages like Tally and Busy record accounting ledger entries, they provide zero workflow support to decide *who to chase, for how much, and why*.

Hisaabset transforms this broken process. It acts as an intelligent decision-support desk that ingests invoices via Gemini AI OCR, automatically isolates disputed amounts from undisputed collectible cash, ranks debtors mathematically based on overdue age and promise reliability, prepares tone-calibrated human-in-the-loop reminders via Amazon SES, and generates verifiable payment settlement receipt vouchers upon payment receipt.

Venture Highlights & Architectural Pillars

Pillar	The Traditional Flaw	The Hisaabset Innovation
Dispute Handling	Retailers use ■2,000 damaged goods to withhold an entire ■50,000 invoice for 60+ days.	**Dispute Isolation:** Automatically segregates ■48,000 collectible cash from ■2,000 held under review.
Debtor Prioritization	Collections staff chase arbitrarily or call favorite friendly accounts, ignoring risky debts.	**Deterministic Scoring:** Multi-factor mathematical formula (Age, Amount, Broken Promises).
Invoice Ingestion	Manual data entry of physical paper/PDF bills leads to data errors and delayed books.	**Multimodal AI Extraction:** Gemini 1.5 Flash extracts line items, totals, and due dates in seconds.
Collections Messaging	Clumsy WhatsApp texts alienate key customers or breach compliance boundaries.	**Human-in-the-Loop SES:** Tone-adaptive (Gentle, Firm, Formal) drafts reviewed before dispatch.
Cloud Architecture	Fragile localhost prototypes that crash or stop when laptops are closed.	**24/7 AWS ECS Fargate:** Production Docker container running continuously in ap-south-1 Mumbai.

1. The Problem Space: The Indian Wholesale Credit Trap

India's wholesale distribution sector powers ■70 Lakh Crore ($850 Billion) of physical commerce. Distributors buy upfront from manufacturers (HUL, ITC, Sun Pharma, Asian Paints) on strict 7-to-15 day credit terms, but are forced by local market dynamics to extend 21-to-45 day unsecured credit to over 15 million neighborhood retail stores (kiranas).

1.1 The Cash Credit (CC) / Bank Overdraft (OD) Debt Trap

To bridge the gap between manufacturer bills and delayed retailer payments, every distributor relies on bank Cash Credit / Overdraft lines at **12% to 14% annual interest**. When ■15 Lakhs of receivables slip into the 60–90 day overdue bucket, the distributor pays **■15,000 to ■17,500 every single month in dead interest costs**, directly obliterating their thin 3%–5% net distribution operating margin.

1.2 The "Dispute Hostage" Phenomenon

In daily distribution, minor friction is inevitable: damaged packaging, expired batches, price scheme differences, or short deliveries. Retailers deliberately exploit these minor discrepancies (representing 2%–5% of invoice value) as leverage to **hold 100% of the payment hostage**. Without a structured dispute-isolation workflow, distributors fail to collect the undisputed 95% cash for months.

> **Industry Truth:** Traditional ERPs (Tally, Busy, Marg) are backward-looking accounting ledgers. They record what happened in the past, but offer zero intelligence on operational prioritization, broken payment promises, dispute ring-fencing, or human-reviewed collection follow-ups.

1.3 Market Sizing: India Addressable Opportunity

Metric	Target Segment & Sizing	Commercial Value
TAM (Total Addressable Market)	All registered GST trade distributors and wholesalers in India (~1.2 Million entities)	**■3,600 Cr / year** ($430M)
SAM (Serviceable Addressable Market)	Organized distributors with ■2 Cr–■50 Cr annual turnover and 100+ retail accounts (~350,000 entities)	**■1,050 Cr / year** ($126M)
SOM (Serviceable Obtainable Market)	Early-adopter FMCG, pharma, and hardware distributors in Tier 1 & 2 industrial hubs (15,000 entities in 3 years)	**■45 Cr / year** ($5.4M)

2. Product Architecture & Core Decision Engines

Hisaabset is architected as an **Air-Gapped Decision Support System**. It provides actionable intelligence to the Finance Lead while ensuring safety by maintaining human-in-the-loop approvals for all external communications and credit decisions.

2.1 Dispute Isolation & Settlement Architecture

When a dispute occurs, Hisaabset partitions the invoice balance into two distinct ledgers: Collectable Balance and Review Amount. If an invoice of ■50,000 has a ■5,000 dispute for damaged goods, the system holds ■5,000 under review and directs collections efforts exclusively toward the safe ■45,000. It supports three dispute resolutions: *Pending Investigation*, *Replacement Goods Delivered*, and *Credit Note Reconciliation*.

2.2 Deterministic Multi-Factor Priority Queue

Unlike heuristic or random follow-ups, Hisaabset ranks overdue collectible debts using a mathematically verifiable scoring algorithm:

```
Base Priority Score:
Score_base = 100 × (0.45 × AgeScore + 0.35 × AmountScore + 0.20 × BrokenPromiseFlag)
Aging Escalation (>90 Days Overdue):
Escalation_factor = 0.25 + 0.50 × min((DaysOverdue - 90) / 90, 1.0)
Final_Priority = Score_base + (100 - Score_base) × Escalation_factor
```

2.3 Multimodal AI Document Ingestion (Google Gemini 1.5 Flash)

Distributors receive physical paper invoices and PDF delivery challans. Hisaabset integrates Google Gemini 1.5 Flash to extract invoice metadata (Invoice Number, Retailer Name, Billed Amount, Line Items, Due Date). Users review extracted data and assign dispute tags directly before saving.

2.4 One-Click Payment Settlement & Printable Receipt Vouchers

When an overdue invoice is settled (via RTGS, NEFT, UPI, or Cash), the Finance Lead enters payment details in the Record Payment module. Hisaabset updates outstanding balances to ■0.00, moves the invoice to Paid status, and generates a formal, printable Payment Settlement Receipt Voucher with unique voucher numbers for retailer verification.

2.5 Human-in-the-Loop Amazon SES Notifications

Hisaabset automatically drafts collection emails calibrated by overdue age: *Gentle Reminder* (1–30 days), *Firm Notice* (31–60 days), and *Formal Demand* (61+ days or broken promises). The Finance Lead reviews and approves the draft before dispatching via Amazon SES with verified DKIM/SPF.

3. Cloud-Native AWS Production Infrastructure

Hisaabset is deployed as a continuous cloud-native containerized application in AWS Region ap-south-1 (Mumbai). It provides 24/7 uninterrupted uptime without local workstation dependencies.

AWS Service	Configuration Details	Engineering Justification
AWS ECS on Fargate	0.5 vCPU, 1 GB RAM, Fargate Linux x86_64 serverless container pod.	Zero server maintenance; automatic OS patching; continuous 24/7 uptime; no cold-start latency.
AWS Elastic Container Registry (ECR)	Private repository: hisaabset-repo with immutable image tags.	Secure, high-bandwidth container image registry integrated directly with ECS task execution roles.
Amazon Simple Email Service (SES)	Production endpoint in ap-south-1; verified sender with in-app sandbox verification.	Enterprise-grade transactional email dispatch with DKIM signing and strict bounce/complaint handling.
AWS VPC & Security Groups	Security Group: hisaabset-ecs-sg; ingress port 8501; unrestricted egress 443.	Network isolation preventing unauthorized access while allowing outbound calls to AWS and Gemini APIs.
AWS IAM Least Privilege	Roles: ecsTaskExecutionRole & task role with targeted SES permissions.	Ensures the container cannot access unauthorized AWS resources outside its operational boundary.
Amazon CloudWatch Logs	Log Group: /ecs/hisaabset-task with 7-day retention policy.	Centralized real-time telemetry, crash monitoring, and audit log tracking for all financial operations.

3.1 Live Deployment Telemetry & Verification

- **Public Endpoint:** http://13.201.115.214:8501
- **Healthcheck Verification:** curl -I http://13.201.115.214:8501/_stcore/health returns HTTP/1.1 200 OK.
- **Container Task:** Active ECS Task Revision hisaabset:7 running in VPC Subnet subnet-09f196924d5cb3673.
- **Security Isolation:** Non-root container user, read-only root filesystem compliance, and no hardcoded secrets.

4. Business Model, AWS TCO & Unit Economics

Hisaabset operates on a high-margin, scalable B2B Software-as-a-Service (SaaS) subscription model designed to match the economic realities of Indian wholesale trade distributors.

4.1 AWS Total Cost of Ownership (TCO) Breakdown

Infrastructure Component	Sizing / Consumption	Monthly Cost (USD)	Monthly Cost (INR)
AWS Fargate Compute (0.5 vCPU)	0.5 vCPU × 730 hours @ $0.04048/vCPU-hr	$14.78	■1,227
AWS Fargate Memory (1 GB RAM)	1.0 GB × 730 hours @ $0.004445/GB-hr	$3.25	■270
AWS ECR Container Storage	2 GB container image @ $0.10/GB	$0.20	■17
Amazon SES Email Dispatch	~2,500 transactional emails/month	$0.25	■21
CloudWatch Logs & Metrics	5 GB ingestion & dashboard telemetry	$2.68	■222
Data Transfer Out & Misc	Outbound Streamlit UI traffic (~10 GB)	$1.00	■83
TOTAL INFRASTRUCTURE RUNNING COST	**Continuous 24/7 Active Pod**	**$22.16 / month**	**■1,840 / month**

4.2 SaaS Pricing Tiers & Packaging

Tier Plan	Target Segment	Price (INR)	Features Included
Starter	Single-category distributors (up to 50 retailers)	**■1,499 / mo** (■14,990/yr)	Balance calculation, dispute logging, PDF exports, standard priority scoring.
Growth / Pro (Core)	Mid-sized FMCG/Pharma (50–250 retailers)	**■2,499 / mo** (■24,990/yr)	All Starter features + Gemini AI PDF Ingestion, Amazon SES email notices, payment vouchers, promise tracking.
Enterprise Multi-Branch	Super-stockists & multi-depot (250+ retailers)	**■4,999 / mo** (■49,990/yr)	All Pro features + Multi-branch consolidation, custom ERP connectors (Tally/Busy), dedicated IP, priority support.

4.3 Unit Economics & Customer ROI Analysis

• **Hosting Scalability:** In a shared container pool, a single 0.5 vCPU Fargate pod easily hosts 30 to 50 active SMB distributors. The direct cloud infrastructure cost per tenant is **under ■65 per month**, yielding an extraordinary **97.4% Gross Margin** at the ■2,499/mo price point.
• **Customer ROI Justification:** For an average distributor carrying ■15 Lakhs in 60-day receivables, accelerating collections by just 14 days saves ■7,000/month in bank overdraft interest and prevents ■8,000/month in write-offs. Against a ■2,499 subscription fee, the distributor earns an immediate **6x to 8x Net ROI**.

5. Engineering Rigor & Testing Validation

Hisaabset was built adhering to strict test-driven development practices. All financial calculations use Python's exact Decimal module, eliminating IEEE 754 floating-point rounding errors.

Test Suite Module	Test File	Count & Status	Scope Covered
Core Loader & Validation	tests/test_loader.py	12 / 12 Passed	Input normalization, date boundaries, null handling, duplicate rejection
Rules & Aging Logic	tests/test_rules.py	26 / 26 Passed	Balances, applied receipts, age buckets, priority ranking formula
Dispute Isolation Engine	tests/test_disputes.py	14 / 14 Passed	Dispute causes, resolution splits, credit note application
Draft Follow-Up & Tone	tests/test_drafts.py	8 / 8 Passed	Customer grouping, tone escalation, broken-promise handling
End-to-End Integration	tests/test_integration.py	10 / 10 Passed	Full starter fixture regression (tests 1–42 specification)
Payments & Vouchers	tests/test_payments.py	9 / 9 Passed	Invoice settlement, status transition, payment voucher generation
Authentication & Role Gate	tests/test_auth.py	6 / 6 Passed	PBKDF2-SHA256 password hashing, session state token validation
Retailer Credit Risk	tests/test_retailer_risk.py	4 / 4 Passed	Repayment history scoring, limit suggestions, break-even validation
TOTAL TEST COVERAGE	**Full Pytest Suite**	**89 / 89 Passed**	**100% Core Business Logic & Mathematical Coverage**

6. Competitive Advantage Matrix

Feature / Capability	Tally / Busy	Khatabook / Vyapar	Hisaabset (LedgerLoop)
Core Purpose	Tax compliance & ledger accounting	Digital bahi-khata for micro-retailers	**Intelligent B2B Collections Decision Desk**
Dispute Isolation	No (binary balance)	No	**Yes (separates disputed from collectable)**
Debtor Prioritization	Chronological aging only	None	**Mathematical Multi-Factor Priority Score**
AI Invoice OCR	Manual entry only	Basic receipt photo capture	**Gemini 1.5 Flash automated extraction**
Human-in-the-Loop Reminders	Manual printouts	Automated WhatsApp spam	**Calibrated tone drafts via Amazon SES**
Payment Settlement Vouchers	Receipt voucher (complex)	None	**Instant one-click printable voucher**
Cloud Availability	Desktop on-premise (Tally Prime)	Mobile-only app	**24/7 AWS ECS Fargate Web Cloud**

7. Hackathon Evidence Verification Checklist (12 / 12)

1. **Public Git Repository:** https://github.com/MdZeeshanAli01/LedgerLoop
2. **Live Hosted Cloud URL:** http://13.201.115.214:8501 (AWS ECS Fargate Mumbai)
3. **40-Character Commit SHA:** 41909b15acb83751a25e296a40981ad7e8547374
4. **Cloud Architecture Evidence:** Documented in docs/TECHNICAL_PRESENTATION_GUIDE.md
5. **Business Pricing & AWS TCO:** Comprehensive spreadsheet & model in docs/BUSINESS_PRICING_AND_AWS_COST_MODEL.md
6. **MBA Presentation Deck Guide:** Slide-by-slide pitch guide in docs/MBA_GUIDE.md
7. **Technical Handoff Document:** Complete specification in docs/TECHNICAL_HANDOFF.md
8. **Automated Test Evidence:** 89 Pytest test suite results in tests/
9. **Sample Data Fixtures:** Starter fixture and edge-case synthetic data in data/
10. **Docker & Deployment Artifacts:** Production container files in deploy/ and app/
11. **SES Notification Verification:** Integrated communication engine in src/communications.py
12. **Invoice Settlement & Vouchers:** Operational closure engine in src/payments.py